

# Lab Report: 6

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Group number: 6

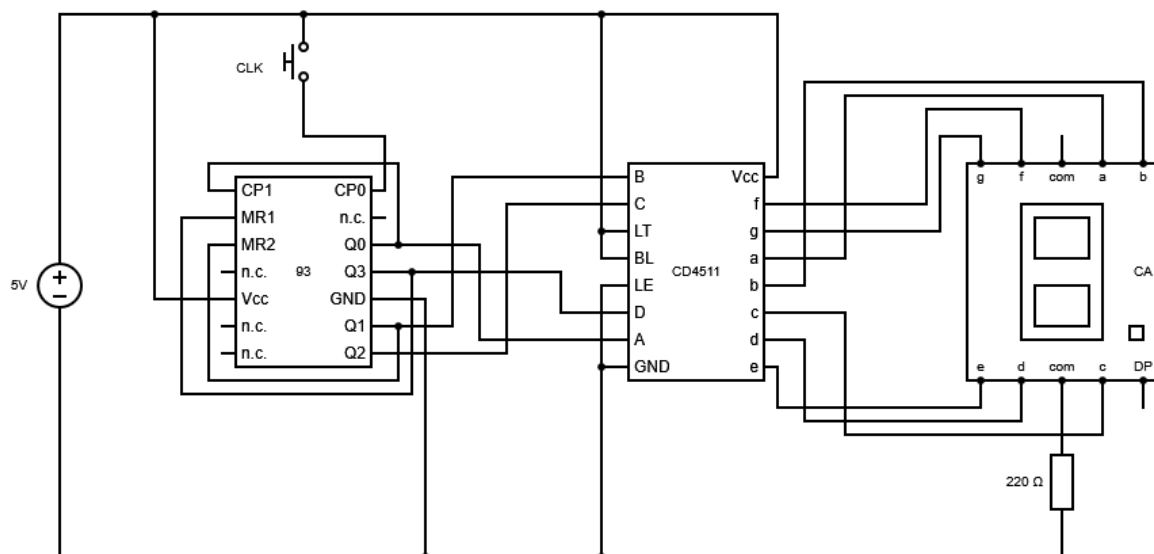
## Experiment-A

**Objective:** Building the circuit for a decade counter

### Electronic components used:

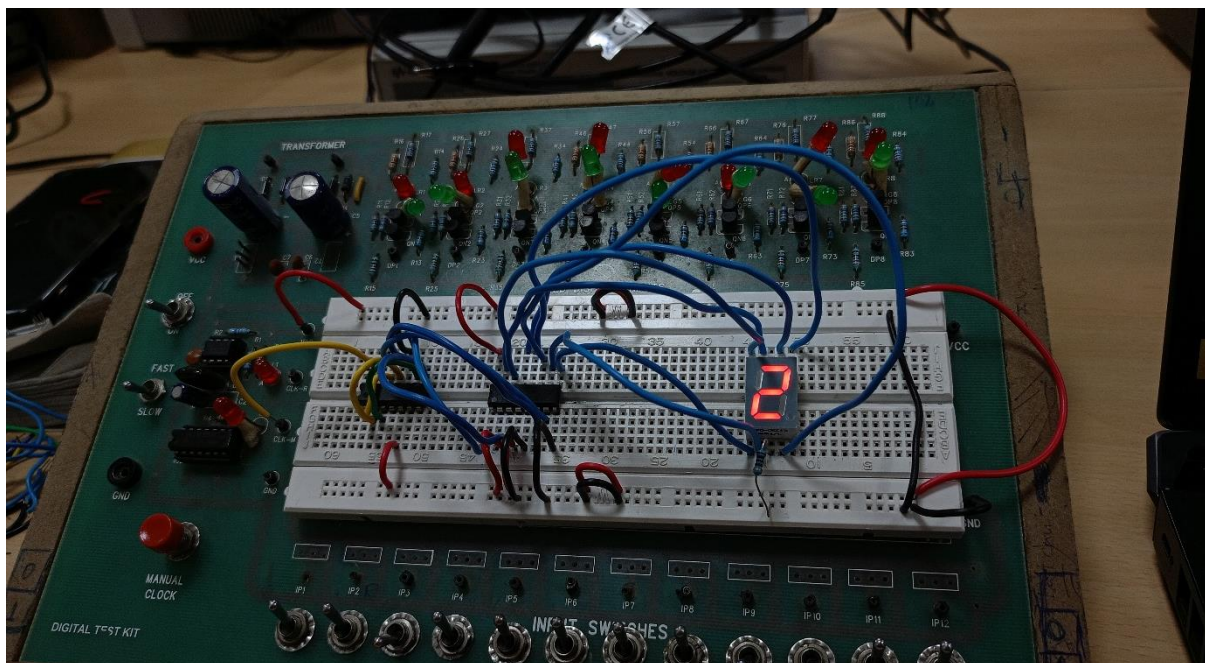
- Digital test kit (w/ breadboard)
- 74HC93 (4-bit ripple counter) IC
- CD4511 (BCD to 7-segment latch decoder) IC
- 7-segment display (Common Cathode)
- Wires

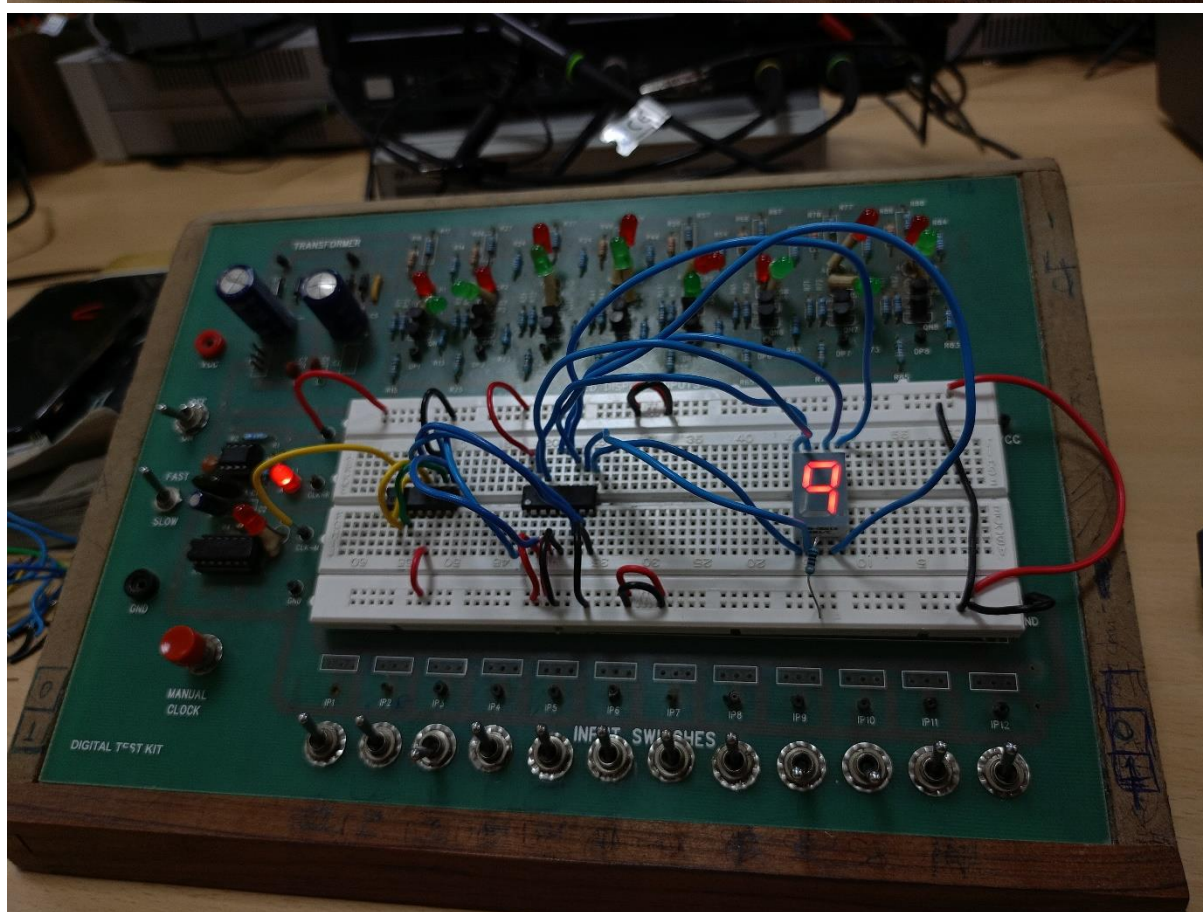
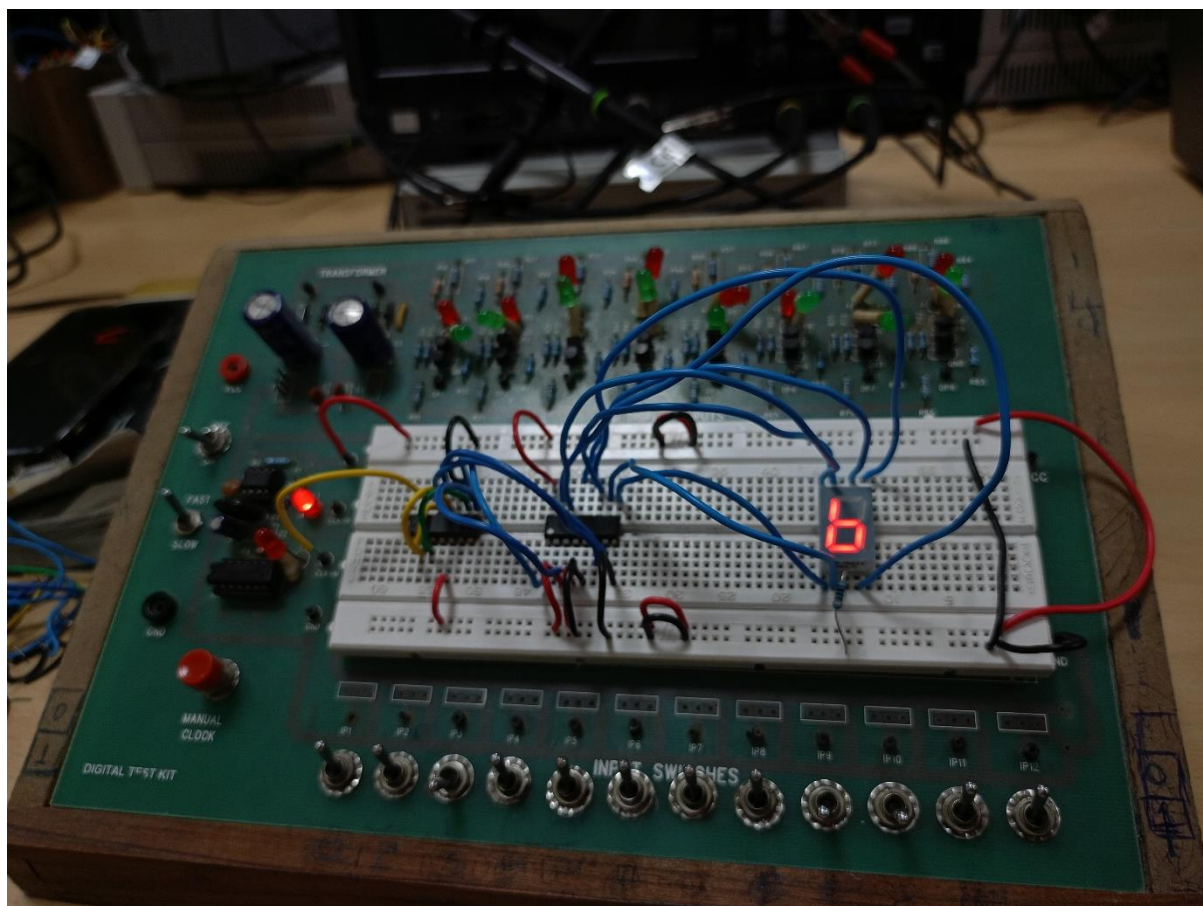
### The reference circuit:



## Procedure:

- Plug in the decoder driver IC (CD4511) and the 7-segment display into the breadboard. Power the IC and connect them as shown in the reference circuit.
- Test the display by setting pin LT (Lamp Test) to GND. Ensure that all the segments are glowing.
- Reset LT to  $V_{CC}$  and plug in the counter IC (74HC93) as well. Connect all the components as shown in the reference diagram.
- Test the circuit by providing clock input through the push button.





## Conclusion:

The decade counter worked as expected, counting up till 9 and then resetting to 0 back again.

Link for the Tinkercad simulation:

<https://www.tinkercad.com/things/8iUQt7w0S5z-parta/editel?sharecode=TM55TRFnMtfMJuWdjfbQlo9cMFi7uVZzH9KMBYkBc64>

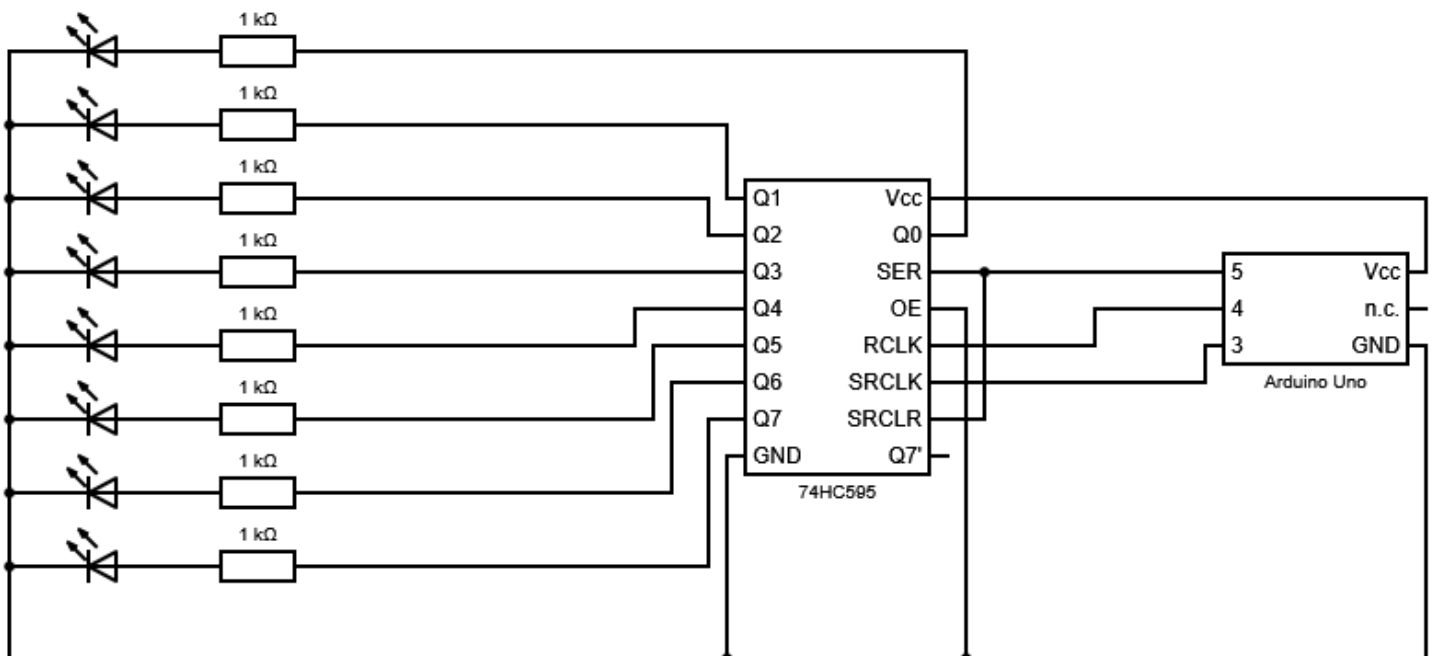
# Experiment-B

**Objective:** Building a shift register circuit and using it with an Arduino

**Electronic components used:**

- Digital test kit (w/ breadboard)
- 74HC595 (8-bit register) IC
- Arduino Uno + USB cable A to B
- Wires

**The reference circuit:**



## Arduino Code:

### Part-A

```
1 //Auth: @Saarthak_2023102055
2 //8-bit counter
3
4 void setup()
5 {
6     for(int i = 3; i <= 5; i++)
7         pinMode(i, OUTPUT);
8 }
9
10 void loop()
11 {
12     for(int i = 0; i < 256; i++) {
13         digitalWrite(4, LOW);
14         shiftOut(5, 3, LSBFIRST, i);
15         digitalWrite(4, HIGH);
16         delay(500);
17     }
18 }
```

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### Part-B

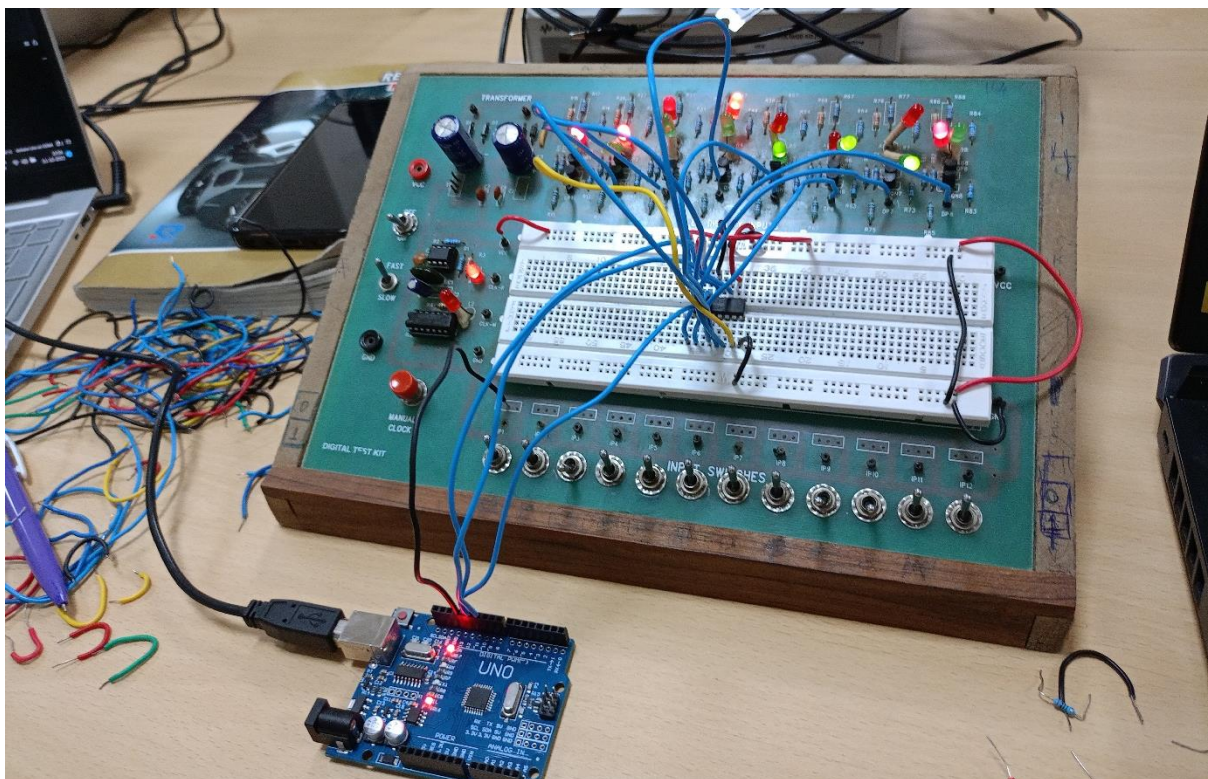
```
1 //Auth: @Saarthak_2023102055
2 //LED controller using shift-register logic
3 //Input: 0-7 (LED no.)
4
5 void setup()
6 {
7     for(int i = 3; i <= 5; i++)
8         pinMode(i, OUTPUT);
9     Serial.begin(9600);
10 }
11
12 void loop()
13 {
14     while(!Serial.available());
15     int i = Serial.parseInt();
16     Serial.print("Glowing LED ");
17     Serial.println(i);
18     digitalWrite(4, LOW);
19     shiftOut(5, 3, LSBFIRST, 1<<i);
20     digitalWrite(4, HIGH);
21 }
```

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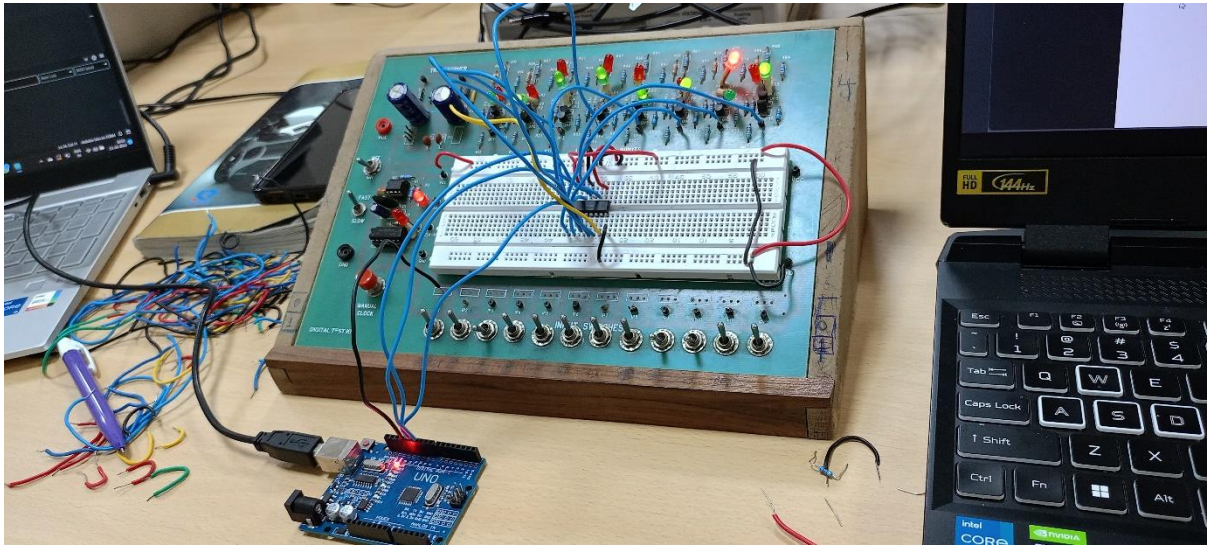
## Procedure:

- Set up the circuit on the breadboard as shown in the reference diagram.
- Upload the code for Part-A to the Arduino.
- Observe the output through the LEDs.
- Now, upload the code for Part-B.
- Provide inputs through the serial monitor and observe the output at the LEDs.

### Part-A



## Part-B



**Conclusion:** An 8-bit shift register circuit was tested successfully.

Link for the Tinkercad simulation:

Part-A:

[https://www.tinkercad.com/things/hZUDV3pa0kT-I6e2parta/editel?sharecode=tGpY3m5YRg8axXvqyZZf049EORhUlrQ18tE22x\\_6yPs](https://www.tinkercad.com/things/hZUDV3pa0kT-I6e2parta/editel?sharecode=tGpY3m5YRg8axXvqyZZf049EORhUlrQ18tE22x_6yPs)

Part-B:

<https://www.tinkercad.com/things/cEQOA8MN1ZO-I6e2partb/editel?sharecode=mDQQP0mQHxkV6IKtgjBiKvMlzjGTzEsFWE3VeweGjYk>